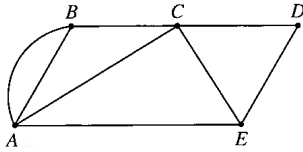


Part A HSC Practice Questions

Networking and Shortest Path

- 1 Consider the network diagram.



What is the sum of the degrees of all the vertices in this network?

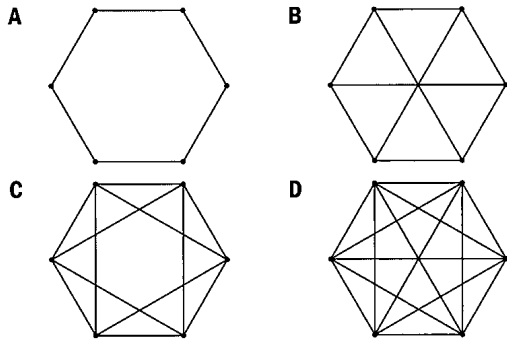
- A 5 B 8
C 14 D 16 (1 mark)

(Q2, 2021 HSC) Medium

- 2 Team A and Team B have entered a chess competition.

Team A and Team B have three members each. Each member of Team A must play each member of Team B once.

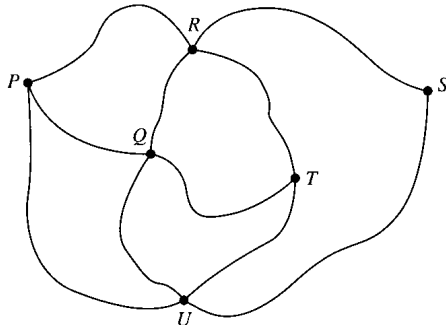
Which of the following network diagrams could represent the chess games to be played?



(1 mark)

(Q9, 2020 HSC) Hard

- 3 A collection of bushwalking tracks is shown on the diagram below. There are six signposts labelled P, Q, R, S, T and U.

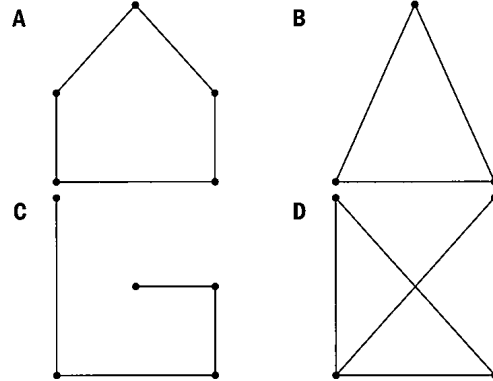


Which of the following edges should be removed so that a bushwalker could walk every track exactly once?

- A QR and QU B SR and SU
C PQ and PU D PQ and QT (1 mark)

Bonus question (see page iv) Medium

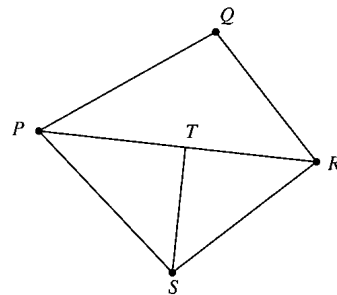
- 4 Which of the following networks has more vertices than edges?



(1 mark)

Bonus question Easy

- 5 Here is a network diagram showing roads between five villages.



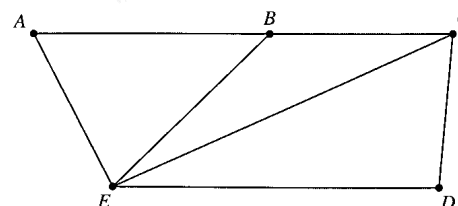
Which of these is a closed path?

- A PQRS B QTRSPQ
C PQRTS D TSRQPT (1 mark)

Bonus question Easy

- 6 Use the table to add arrows and weights to the network diagram.

		To				
		A	B	C	D	E
From	A	—	6	—	—	7
	B	—	—	5	—	—
	C	—	—	—	9	4
	D	—	—	—	—	—
	E	—	8	—	3	—



(2 marks)

Bonus question Easy

Networking and Shortest Path

- 7 Draw a network diagram to represent the roads between towns A, B, C, D and E if:

- Town A is 21 km from B and 16 km from E
- Town B is 38 km from C and 24 km to D
- Town C is 29 km from E and 9 km from D .

(1 mark)

Bonus question Easy

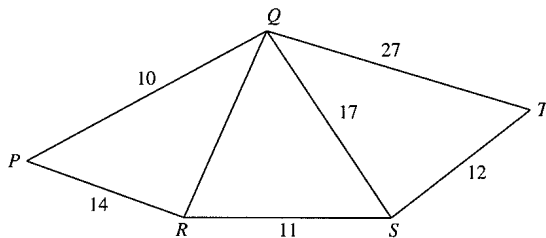
- 8 Three airlines are named according to their base city and can only fly from their base airport to and from two other airports. Airline A flies to and from B and E , C flies to and from A, B and E , and D flies to and from C and E . The cost of a ticket for each airline is the same for any of its routes. Airline A charges \$120, C charges \$160 and D charges \$150.

Draw a weighted network diagram to represent the flight routes and costs.

(2 marks) Easy

Bonus question

- 9 Aleece drove from P , through Q and R , and returned to P . She had travelled a total of 36 km.



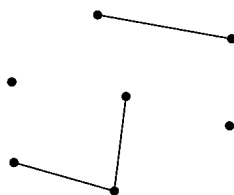
Her friend Gladys left R , drove through Q, T and S , and arrived back in R . How far did Gladys drive?

- A 50 km B 60 km
C 62 km D 65 km

(1 mark)

Bonus question Easy

- 10 Consider the diagram below.



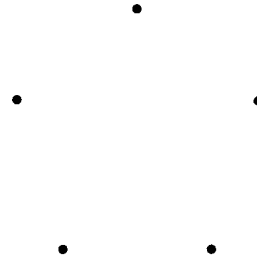
What is the smallest number of additional edges needed to make this a connected network?

- A 1 B 2
C 3 D 4

(1 mark)

Bonus question Easy

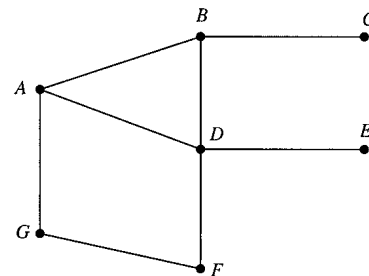
- 11 An undirected connected network has five vertices. Three of these vertices are of even degree and the remaining of odd degree. Show the edges on the diagram below.



(1 mark)

Bonus question Easy

- 12 In the network below, how many vertices have an even degree?

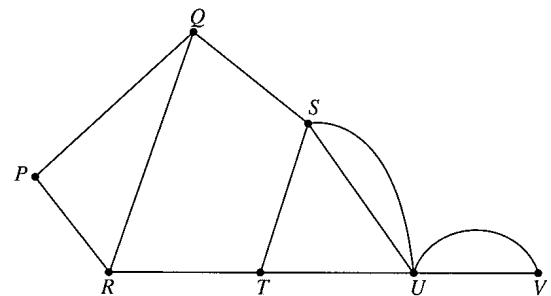


- A 1 B 2
C 3 D 4

(1 mark)

Bonus question Easy

- 13 What is the most common degree of the vertices in the network?



- A 2 B 3
C 4 D 5

(1 mark)

Bonus question Easy

- 14 Brianna drew a network which has n edges. Which expression gives the sum of the degrees of the vertices in the network?

- A n B $2n$
C $n + 1$ D $2n - 1$

(1 mark)

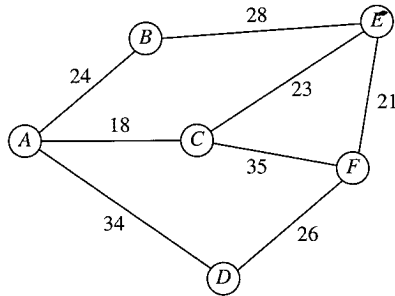
Bonus question Medium

- 15 The network diagram below shows the distances between six villages.

Part A HSC Practice Questions

Networking and Shortest Path

- 15 The network diagram below shows the distances between six villages.



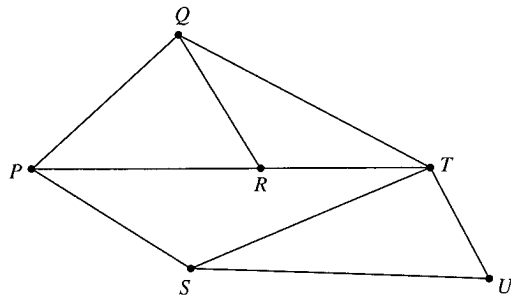
Complete the table representing this network.

	A	B	C	D	E	F
A	—					
B		—				
C			—			
D				—		
E					—	
F						—

(1 mark)

Bonus question Easy

- 16 Which of the following represents a cycle in the network diagram?

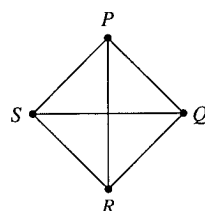


- A PSTUSP B QRTUSP
C TUSPQ D QPSTQ

(1 mark)

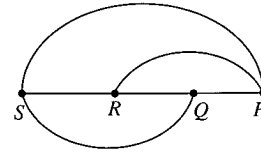
Bonus question Easy

- 17 The diagram shows a network of four points P, Q, R and S.

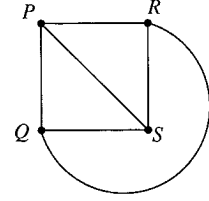


Which of the following network diagrams is **not** an equivalent graph of the network above?

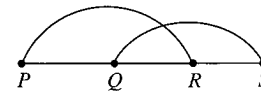
A



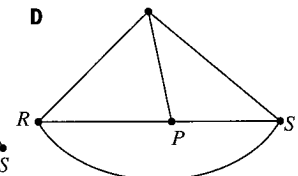
B



C



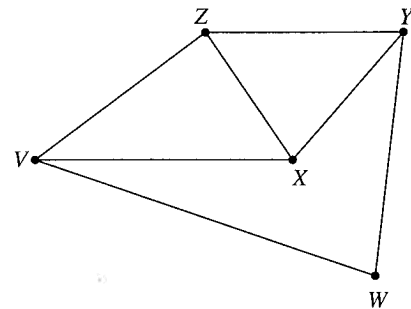
D



(1 mark)

Bonus question Easy

- 18 Which of the following walks is a path in the network diagram above?



- A VZYW B VZYWV
C ZYXZV D XYXVZ

(1 mark)

Bonus question Easy

- 19 The table below represents a directed network.

		To			
		P	Q	R	S
From	P	—	6	—	4
	Q	2	—	—	7
	R	—	5	—	—
	S	—	—	3	—

Complete the network showing weights and arrows.

P •

• Q

S •

• R

(2 marks)

Bonus question Easy

Networking and Shortest Path

- 20 Here is a list of people who invited friends to their birthday parties this year.

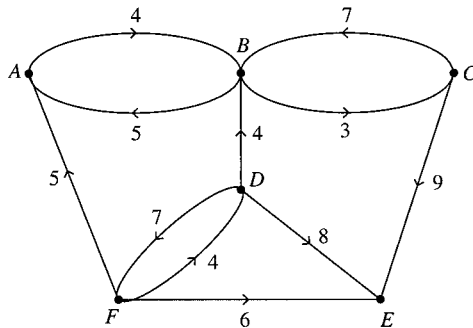
Person	Alex	Beck	Cael	Dana	Erin
Friends	Beck, Dana, Erin	Alex, Erin	Dana, Erin	Alex, Cael, Erin	Alex, Beck, Cael, Dana

Using the letters A, B, C, D and E, represent this information on a network.

• •
• • • (1 mark)

Bonus question Easy

- 21 A directed network is shown below.

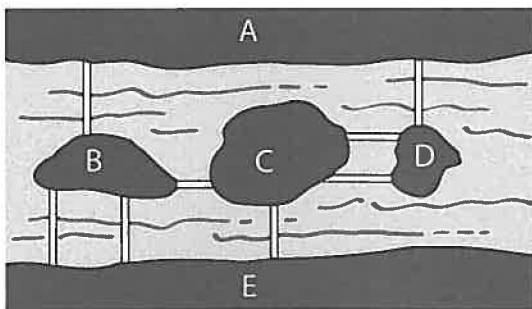


Complete the table network showing weights and arrows.

		To					
		A	B	C	D	E	F
From	A	-					
	B		-				
	C			-			
	D				-		
	E					-	
	F						-

Bonus question Easy (2 marks)

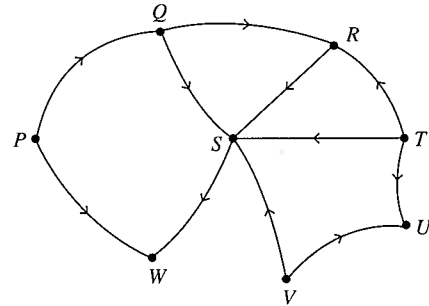
- 22 The map shows a wide river with three islands B, C and D situated in the river. Bridges are shown connecting the banks of the river (A and E) to the islands.



Draw a network diagram to show the road system linking the three islands to the riverbanks.

Bonus question Easy (1 mark)

- 23 The directed network represents a series of one-way streets. The vertices represent the intersection of these streets.

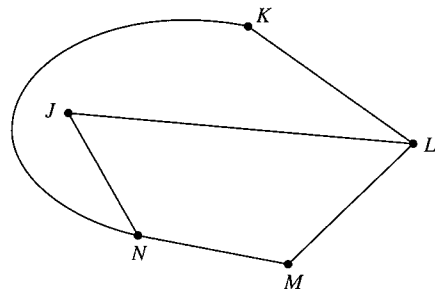


How many vertices can be reached from P?

- A 7 B 4
C 3 D 5 (1 mark)

Bonus question Easy

- 24 A rugby league team welcomed five new players to the squad for the new season. The players were Joey, Kirk, Leon, Matt and Noah. The network diagram shows the players who had played together in other teams over previous seasons.



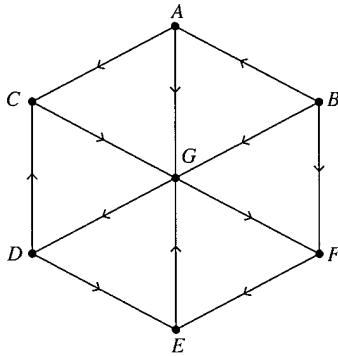
- a How many of these players had Leon played with before joining the team? (1 mark) Easy
b Who had played with Noah before joining the team? (1 mark) Easy
c During the season Owen joined the team. He had played with Joey and Leon in previous seasons. Represent this information on the diagram above. (1 mark) Easy

Bonus question

Part A HSC Practice Questions

Networking and Shortest Path

- 25 The network diagram shows the directional traffic between seven rooms in an art gallery exhibition.



Between which two rooms is it not permitted to move?

- A** A to E **B** B to E
C D to B **D** C to E (1 mark)

Bonus question Easy

- 26 The table shows the distances in kilometres between towns A to E.

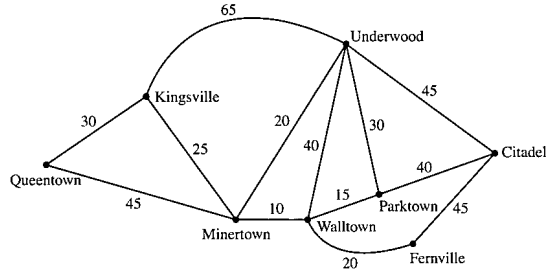
	A	B	C	D	E
A	–	42	64	–	53
B	42	–	31	–	–
C	64	31	–	51	65
D	–	–	51	–	48
E	53	–	65	48	–

Draw a network diagram to represent the information in the table. (2 marks)

Bonus question Easy

Networking and Shortest Path

- 1 The network diagram shows the travel times in minutes along roads connecting a number of different towns.



- a Draw a minimum spanning tree for this network and determine its length.

(3 marks) **Easy**

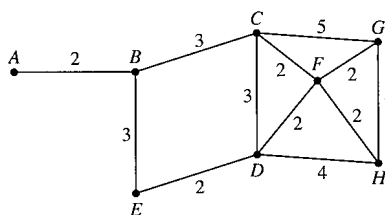
Length of minimum spanning tree =

- b How long does it take to travel from Queentown to Underwood using the fastest route?

(1 mark) **Medium**

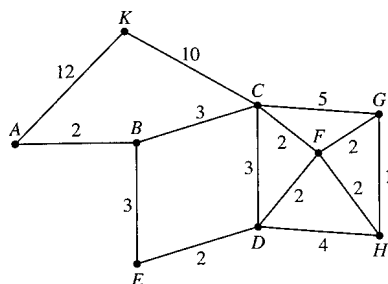
(Q23, 2021 HSC)

- 2 The diagram represents a network with weighted edges.



- a Draw a minimum spanning tree for this network in the space below and determine its length.
- b The network is revised by adding another vertex, K. Edges AK and CK have weights of 12 and 10 respectively, as shown.

(3 marks) **Medium**

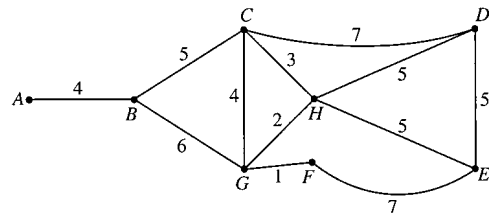


What is the length of the minimum spanning tree for this revised network?

(1 mark) **Medium**

(Q18, 2020 HSC)

- 3 The network diagram shows the tracks connecting 8 picnic sites in a nature park. The vertices A to H represent the picnic sites. The weights on the edges represent the distances along the tracks between the picnic sites, in kilometres.



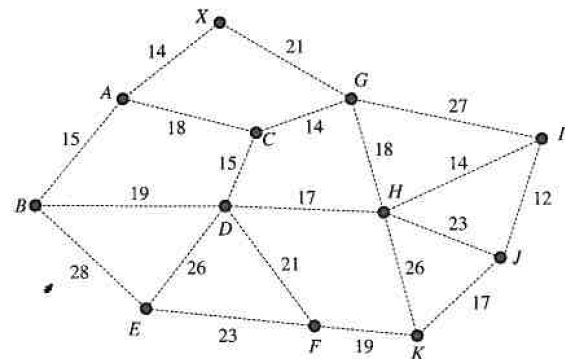
- a Each picnic site needs to provide drinking water. The main water source is at site A. By drawing a minimum spanning tree in the space below, calculate the minimum length of water pipes required to supply water to all the sites if the water pipes can only be laid along the tracks.
- b One day, the track between C and H is closed. State the vertices that identify the shortest path from C to E that avoids the closed track.

(2 marks) **Medium**

(1 mark) **Medium**

(Q30, 2019 HSC)

- 4 A market gardener is expanding his cropping area and wants to place taps at locations A to K on the diagram below. Each tap is to be connected with a water pipe from the existing tap X. The distance, in metres, from each adjacent tap is recorded.



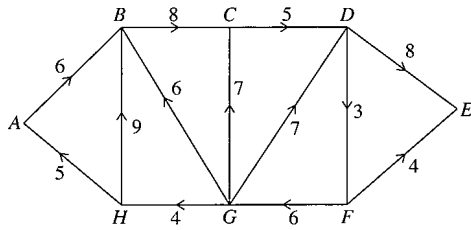
On the diagram draw the minimum spanning tree required to do this and determine the total cost if each new tap costs \$22 and the cost of laying pipe is \$4.50/metre, in metres.

(2 marks)

Bonus question (see page iv) **Hard**

Networking and Shortest Path

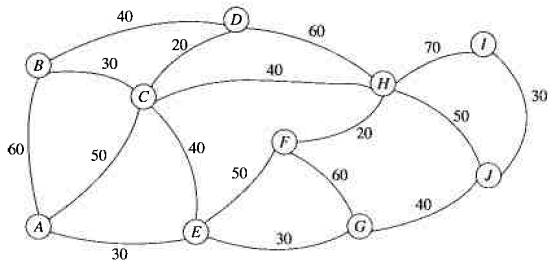
- 5 The diagram shows a directed network where the weights are in minutes.



- a From which vertex can vertex A not be reached? (1 mark) **Easy**
- b What is the minimum time to travel from A to G? (1 mark) **Medium**

Bonus question

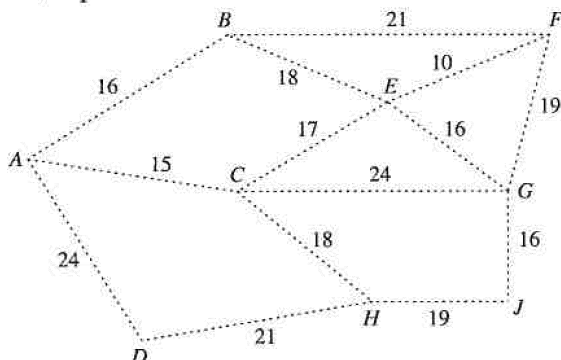
- 6 Once upon a time in the small village of Murgup the pathways between its 10 houses always became very muddy in wet weather. The villagers decided to pave some of the paths with cobblestones so that all householders could travel to everybody else's house on cobblestones without walking in mud.



The distance between the houses is shown on the diagram, in metres, and 60 cobblestones are used for every metre of road. If every cobblestone has an average mass of 5 kg, find the minimum mass of cobblestones required to complete the project. (3 marks)

Bonus question **Hard**

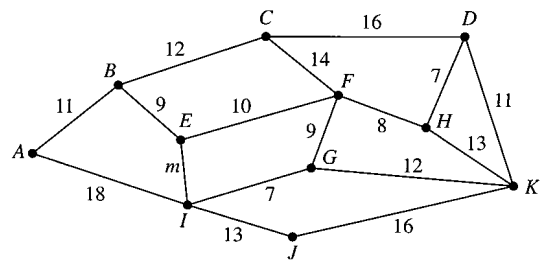
- 7 A computer network is to be installed, consisting of nine computers which need to be connected to at least one other computer by cable. All dimensions are in metres and the cable and installation costs are \$35 per metre.



- a Draw the minimum spanning tree on the network diagram and calculate the total cost of cable required. (1 mark) **Medium**
- b Before work was to have commenced, it was decided that a computer should be removed from the network. What will be the savings if computer A is removed from the minimum spanning tree? (2 marks) **Hard**

Bonus question

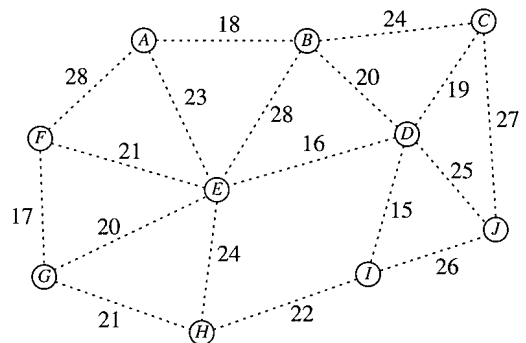
- 8 The minimum spanning tree for the network below includes the edge with weight labelled m .



Find the value of m , if the total weight of the minimum spanning tree is 97. (1 mark)

Bonus question **Medium**

- 9 Andrew manages a holiday park and a contractor has quoted \$42 per metre to provide connecting paths to each of the cabins as shown in the network diagram below. All measurements are in metres.

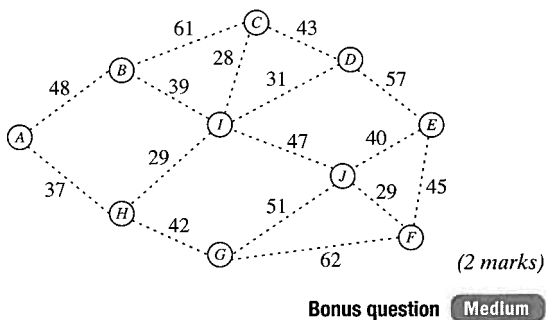


What will be the minimum cost of providing at least one path to each cabin in the network? (2 marks)

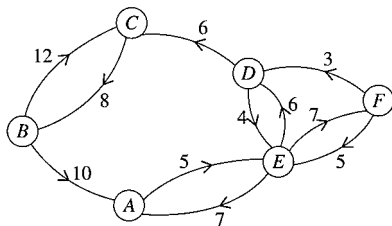
Bonus question **Hard**

- 10 The network diagram shows the locations of 10 rides at a theme park. The distance in metres between the rides is shown on the diagram. Draw the minimum spanning tree for the network on the diagram and write the length.

Networking and Shortest Path



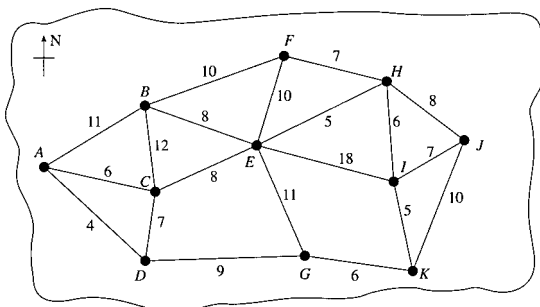
- 11 The network diagram shows the bus routes between stops in a certain town, and the time in minutes to travel between stops.



- a What is the quickest time to travel between E and B? (1 mark) Medium
- b In how many different ways is it possible for a person to travel from F to A without passing through a bus stop more than once? Explain your answer. (1 mark) Hard

Bonus question

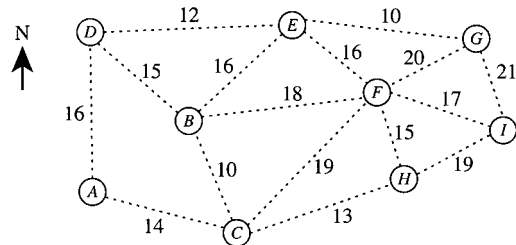
- 12 The map shows the distances, in metres, between 11 new statues in a park shown by the letters A to K. As the statues are to be floodlit at night, electricity is to be provided to the base of each statue. The cost of laying the electricity cable is \$42/metre.



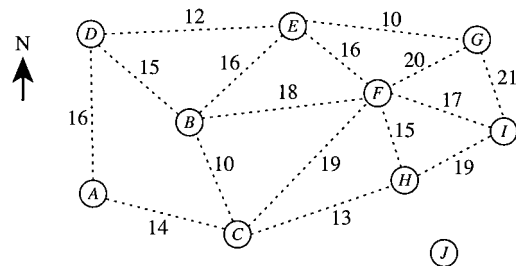
- a What is the minimum cost of laying the cable that joins each of the statue locations? (2 marks) Medium
- b Before work is to proceed, the project manager considers not running the cable to the statue at H to save money. Discuss the feasibility of the decision, showing calculations. (2 marks) Hard

Bonus question

- 13 Arnie has a small acreage and is planting trees across a paddock. He plans to lay water pipes to eight spots where he will locate taps. An existing tap is already located at C. The distance, in metres, between each tap is shown on the network diagram below.



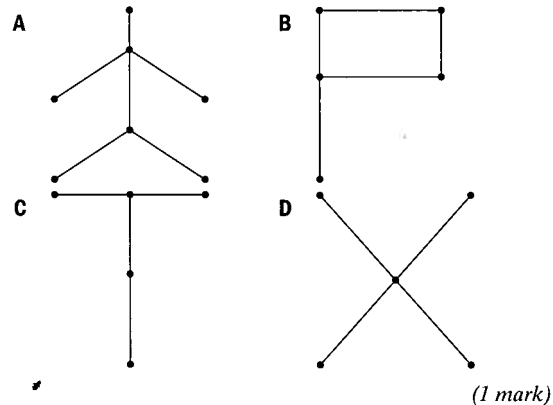
- a On the diagram draw the minimum spanning tree required to do this and determine the total length of pipe required, in metres. (2 marks) Medium
- b Before he starts, Arnie decides to include another tap at location J which is shown on the network diagram below, where $JC = 12$ m, $JH = 10$ m and $IJ = 16$ m.



Determine whether the inclusion of the new tap will increase the total length of pipe required. Support your argument with calculations. (2 marks) Hard

Bonus question

- 14 Which of the following network diagrams is not a tree?

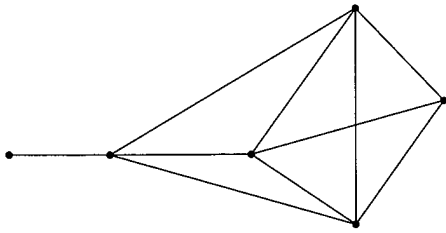


Bonus question Easy

Part B HSC Practice Questions

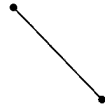
Networking and Shortest Path

15



Which of the following is a spanning tree of the network above?

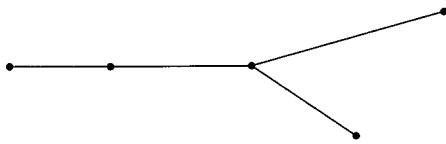
A



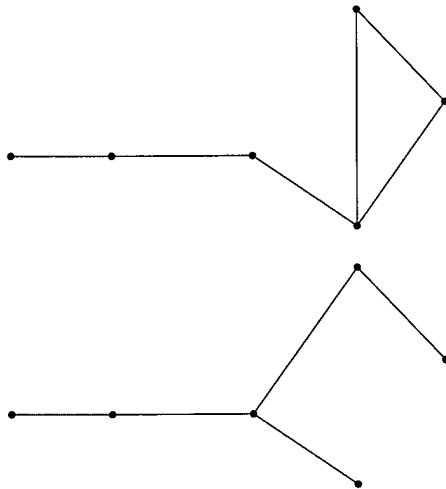
B



C



D



Bonus question Easy

16 The table below is used to represent a network.

	A	B	C	D	E
A	—	6	5	14	12
B	6	—	—	9	—
C	5	—	—	8	7
D	14	9	8	—	4
E	12	—	7	4	—

a Draw a weighted network that is represented by the given table.

A.

C.

E.

B.

D.

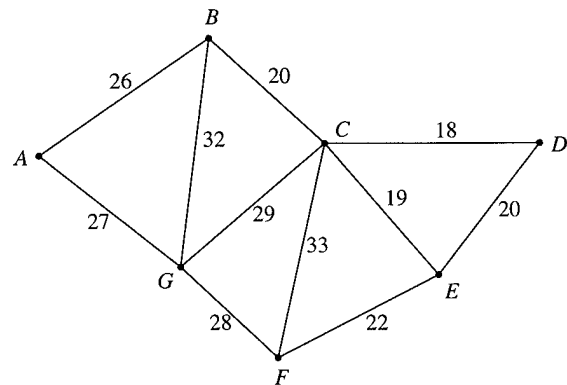
(2 marks) Easy

b Identify the minimum spanning tree on your network diagram above. What is the length of the minimum spanning tree?

(2 marks) Easy

Bonus question

17 The network below shows the road distances in kilometres between seven towns labelled A, B, C, D, E, F and G. To establish a cable network of communications, it is proposed to lay the underground cable besides existing roads.



a By highlighting or emboldening show the minimum spanning tree on the network diagram that ensures all the towns are connected to the network and minimises the total amount of cable used.

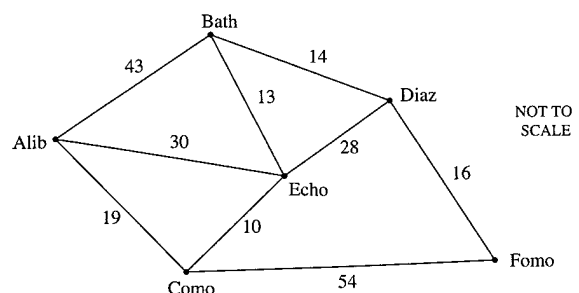
(1 mark) Medium

b A company quoted 4.1 million dollars to lay the cable. If it costs \$26.50/metre to lay the cable, what will be the amount of profit made by the company?

(2 marks) Medium

Bonus question

18 The network diagram shows the roads joining six towns. All measurements are in kilometres.



NOT TO SCALE

Part B HSC Practice Questions

Networking and Shortest Path

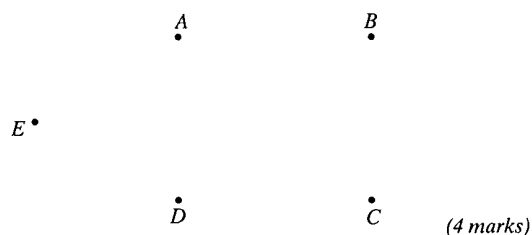
What is the shortest possible route between Alib and Fomo?

- A 69 km B 70 km
C 72 km D 73 km (1 mark)

Bonus question Easy

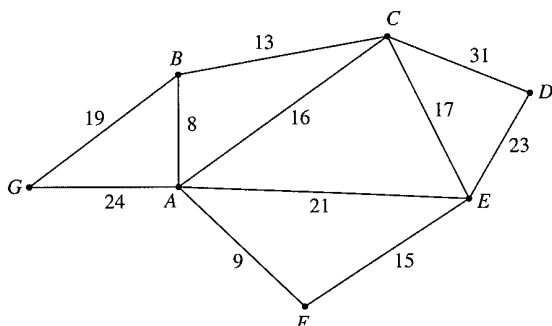
- 19 A university residential college has completed a study to determine the cost of installing a new fibre-optic network on the campus. The cost of joining the five main buildings is provided in the table below, expressed in thousands of dollars. Buildings that cannot be joined are shown as a '-'. Using a minimum spanning tree, find the cost of connecting all five buildings.

	A	B	C	D	E
A	-	14	21	17	11
B		-	11	-	12
C			-	10	16
D				-	13
E					-



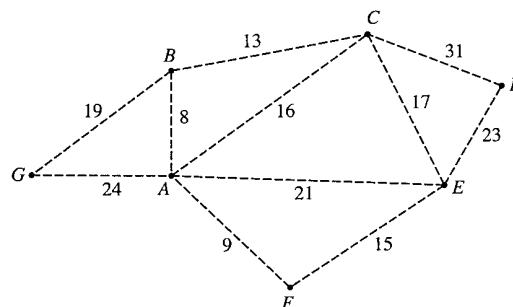
Bonus question Medium

- 20 The network diagram below shows a small park with six garden beds, labelled A, B, C, D, E and F. The park is accessed through a gate shown as G. All dimensions are in metres.



- a What is the minimum distance a person needs to walk to visit each garden bed, starting and finishing at the gate? (1 mark) Easy
- b Water is available at the entrance of the park. New pipes are to be laid along the paths to provide water to each of the garden beds. Find the minimum length of

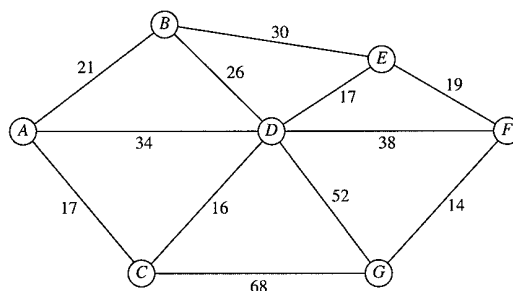
pipes required. Support your answer by drawing a minimum spanning tree on the diagram below.



(2 marks) Medium

Bonus question

- 21 The network diagram shows the average time taken, in minutes, to walk between seven villages using existing tracks.



What is the shortest possible time to travel between A and G? Describe the path. (2 marks)

Bonus question Medium

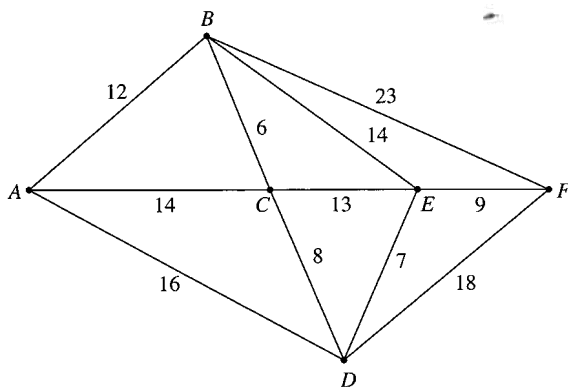
- 22 A network contains n vertices. How many edges are required to form a spanning tree?
- A n B $n - 1$
C $n + 1$ D $2n$ (1 mark)

Bonus question Easy

- 23 The diagram below shows a train network through six towns, Ashford, Bellbird, Cudal, Dungog, Euston and Falco.

The numbers on each edge represent the cost (in dollars) of an adult ticket between the two towns. Jannah and her eight-year-old daughter Ella live in Ashford and will be travelling by train to visit a friend who lives in Falco. The price of a child ticket is 75% of an adult ticket.

Networking and Shortest Path

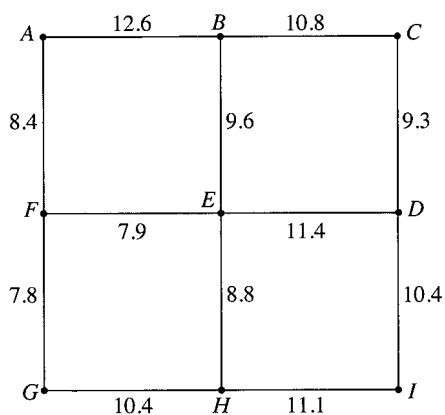


What is the minimum cost of the two tickets?
(2 marks)

Bonus question **Medium**

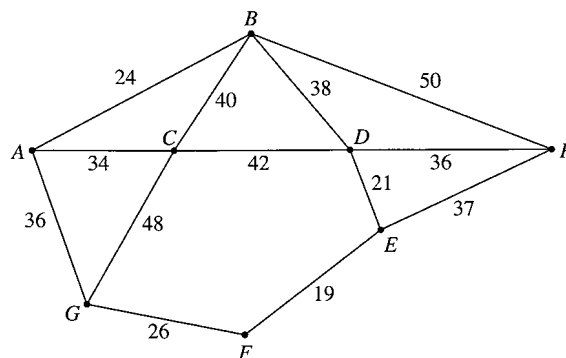
- 24** The following network shows the length, in kilometres, of roads connecting nine towns. The local council plans to resurface some of the roads to ensure that people living in each town can access all other towns along a resurfaced road.

What length of road will connect all nine towns while keeping the length of road to be resurfaced to a minimum?



- A** 73.0 km **B** 74.6 km
C 75.0 km **D** 76.0 km (1 mark)

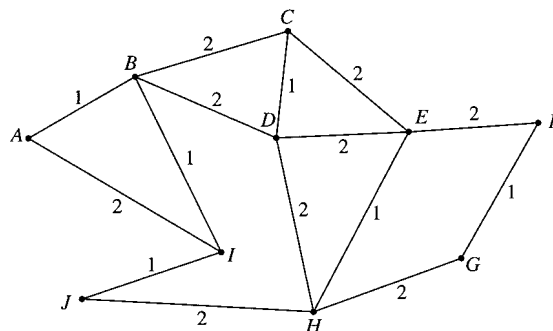
Bonus question **Medium**



The successful contractor has quoted a rate of \$18.40 per metre. What would be the minimum cost of the project? (3 marks)

Bonus question **Medium**

- 26 The network diagram below represents a local fitness park which has 10 exercise stations named A to J . The edges on the graph represent the tracks between each station.



Logan starts at station *C* and runs every track exactly once. He rests for 5 minutes and then runs the shortest route back to *C*. What is the total distance covered? (2 marks)

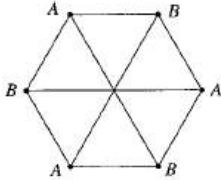
Bonus question **Medium**

- 25 Cabins in a national park camping ground are to be connected to a power source situated at P . The graph below shows the distances, in metres, between seven cabins and P .

Networking and Shortest Path

- 1 Sum of degrees
 $= 4 + 3 + 4 + 2 + 3$
 $= 16$
 Answer D

- 2 Each member of Team A plays each member of team B.



Answer B

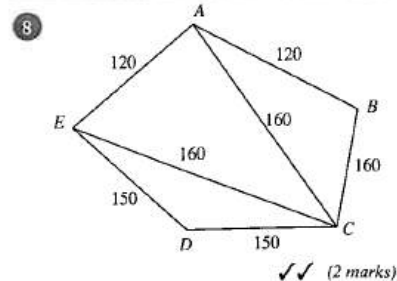
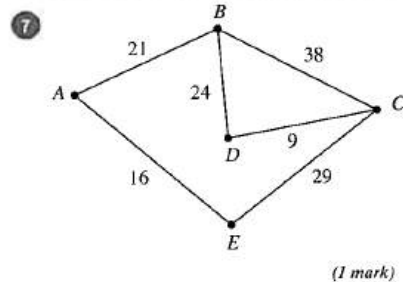
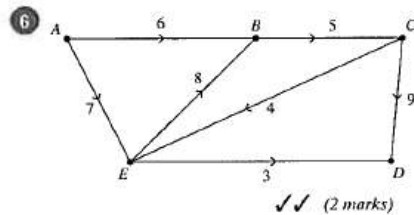
- 3 Each vertex should have an even degree. Removing PQ and QT will mean P, Q and T are each degree 2.
 Answer D

- 4 The network has 5 vertices and 4 edges.



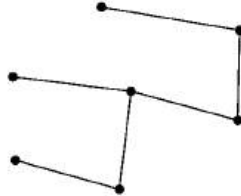
Answer C

- 5 The closed path is TSRQPT.
 Answer D

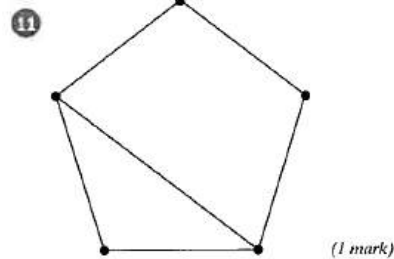


- 9 For PQR: $10 + QR + 14 = 36$
 $QR + 24 = 36$
 $QR = 12$
 For RQTSR: $12 + 27 + 12 + 11 = 62$
 Gladys drove 62 km.
 Answer C

- 10 Another 3 edges are required.



Answer C



- 12 A is degree 3, B(3), C(1), D(4), E(1), F(2) and G(2).
 There are 3 vertices which have an even degree.
 Answer C

- 13 P is degree 2, Q(3), R(3), S(4), T(3), U(5) and V(2).
 The most common degree is 3.
 Answer B

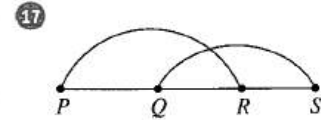
- 14 The sum of the degrees of the vertices of the network is twice the number of edges.
 Answer B

15

	A	B	C	D	E	F
A	—	24	18	34	—	—
B		—	—	—	28	—
C			—	—	23	35
D				—	—	26
E					—	21
F						—

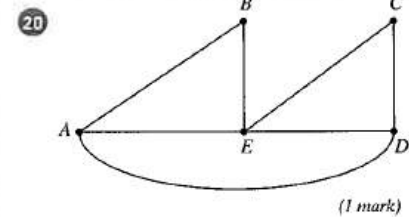
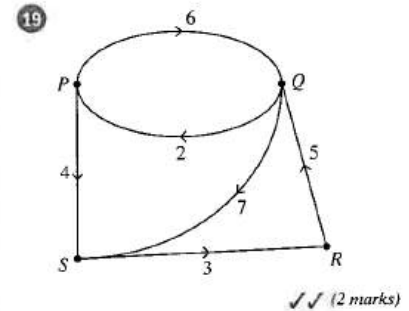
(1 mark)

- 16 A cycle is a walk where the initial and final vertices are the same, but no other vertices are repeated. QPSTQ is a cycle. (A cycle is a closed path.)
 Answer D



Answer C

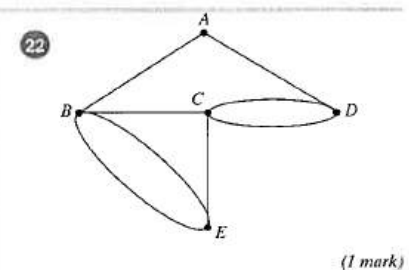
- 18 A path is a walk which never visits the same vertex more than once. VZYW is a path.
 Answer A



21

		To					
		A	B	C	D	E	F
From	A	—	4	—	—	—	—
	B	5	—	3	—	—	—
	C	—	7	—	—	9	—
	D	—	4	—	—	8	7
	E	—	—	—	—	—	—
	F	5	—	—	4	6	—

✓✓ (2 marks)



- 23 From P, the vertices that can be reached are Q, S, W and R. There are 4 vertices.
 Answer B

- 24 a 3
 b Joey, Kirk and Matt
- (1 mark)
 (1 mark)

Part B

(1 mark)

Answer C

✓✓ (2 marks)

Length of minimum spanning tree = 160 min

✓✓✓ (3 marks)

$$\begin{aligned}\text{Time} &= (45 + 20) \text{ min} \\ &= 65 \text{ min} \\ &= 1 \text{ h } 5 \text{ min}\end{aligned}$$

(1 mark)

✓✓

[*BE* could be joined instead of *BC*; *FH* instead of *FG*]

(3 marks)

Minimum length = $14 + 10 = 24$

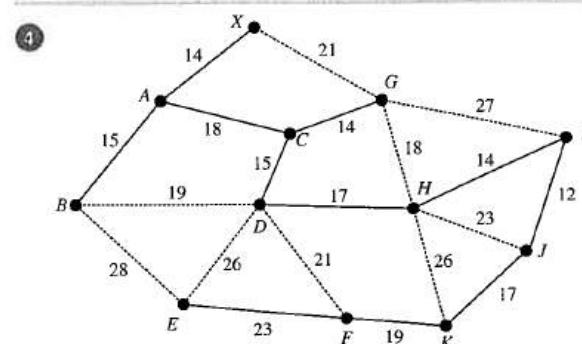
(1 mark)

✓

Minimum length = 25 km

(2 marks)

(1 mark)



2

Networking and Shortest Path

$$14 + 15 + 18 + 14 + 15 + 17 + 14 + 12 + 17 + 19 + 23 = 178$$

∴ the length is 178 m.

$$\therefore \text{Cost} = 178 \times 4.50 + 11 \times 22$$

$$= 1043$$

∴ the total cost is \$1043

(2 marks)

- 5 a A cannot be reached from E.

(1 mark)

- b The path is *ABCDFG* which is
 $6 + 8 + 5 + 3 + 6 = 28$ minutes.

(1 mark)

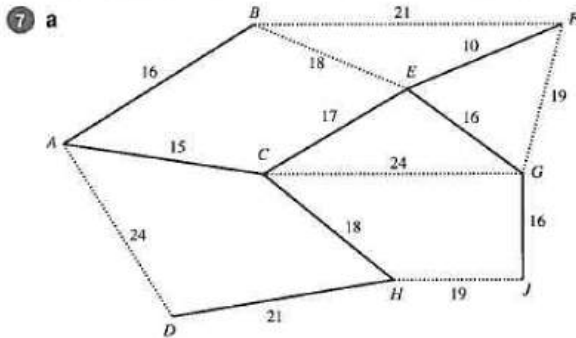
- 6 Distance = $30 + 30 + 20 + 40 + 40 + 30 + 20 + 40 + 30$
 $= 280$

$$\text{Mass of cobblestones} = 280 \times 60 \times 5$$

$$= 84000$$

∴ the mass is 84000 kg, or 84 tonnes.

(3 marks)



$$\text{Length} = 16 + 15 + 17 + 10 + 16 + 16 + 18 + 21$$

$$= 129$$

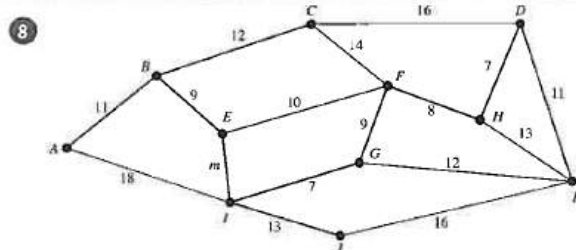
$$\text{Cost} = 129 \times \$35$$

$$= \$4515$$

(1 mark)

- b Reduction in length
 $= 15 + 16 - 18$
 $= 13$
 Savings = $13 \times \$35$
 $= \$455$

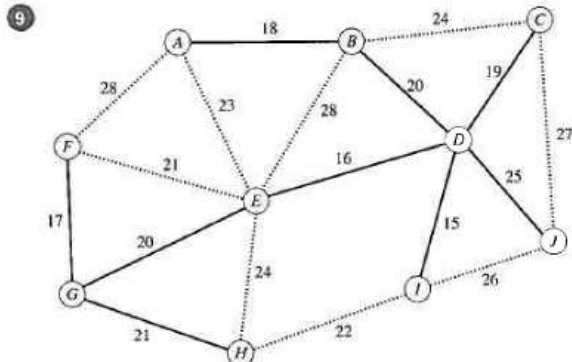
(2 marks)



$$11 + 9 + m + 12 + 7 + 13 + 9 + 8 + 7 + 11 = 97$$

$$\therefore m = 10$$

(1 mark)



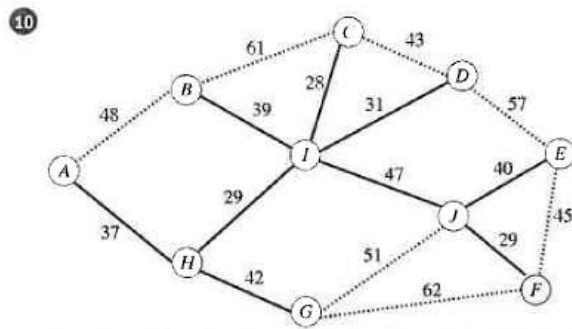
$$\text{Length} = 17 + 20 + 21 + 18 + 16 + 20 + 19 + 15 + 25$$

$$= 171$$

$$\text{Cost} = 171 \times \$42 = \$7182$$

∴ the cost is \$7182.

(2 marks)



$$\text{Length} = 37 + 29 + 42 + 39 + 28 + 31 + 47 + 40 + 29$$

$$= 322$$

(2 marks)

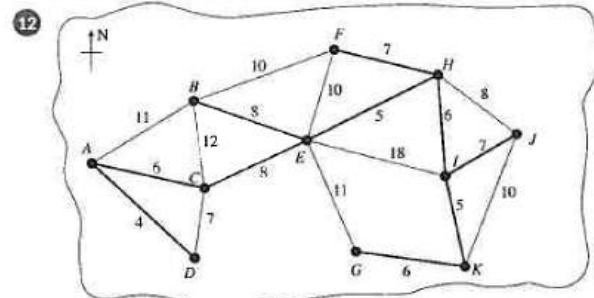
- 11 a *EDCB*: $6 + 6 + 8 = 20$

∴ the quickest time is 20 minutes.

(1 mark)

- b The paths are *FEA*, *FDEA*, *FDCBA*.
 ∴ there are 3 ways.

(1 mark)



- a Length of minimum spanning tree
 $= 6 + 4 + 8 + 6 + 5 + 7 + 6 + 5 + 7 + 8$
 $= 62$
 Cost = 62×42
 $= 2604$
 ∴ the cost is \$2604.

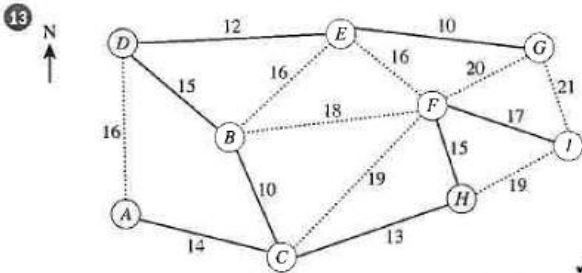
(2 marks)

- b Length of new minimum spanning tree
 $= 6 + 4 + 9 + 6 + 5 + 7 + 8 + 8 + 10$
 $= 63$
 Cost = 63×42
 $= 2646$
 ∴ the cost is \$2646.
 Increase = \$42

∴ even though there is one less statue involved,
 the price increases by \$42.

(2 marks)

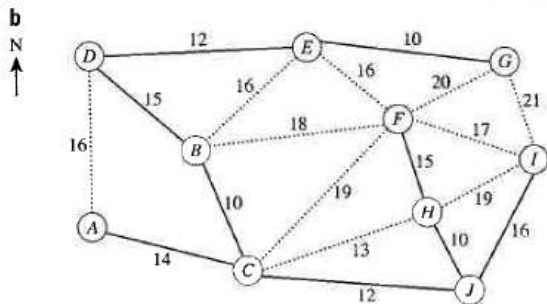
Networking and Shortest Path



a Length = $14 + 10 + 15 + 12 + 10 + 17 + 15 + 13$
 $= 106$

$\therefore$ 106 metres of water pipe

(2 marks)



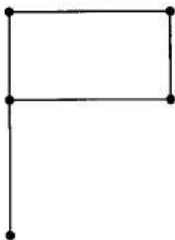
New length = $14 + 10 + 15 + 12 + 10 + 15 + 10$
 $+ 12 + 16$
 $= 114$

Difference = $114 - 106$
 $= 8$

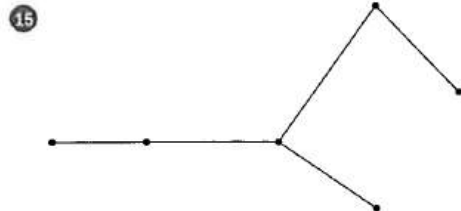
$\therefore$ Yes, the length will increase. Arnie needs another 8 m of water pipe.

(2 marks)

- 14 This network has vertices that are connected by more than one path. This means the network is not a tree.

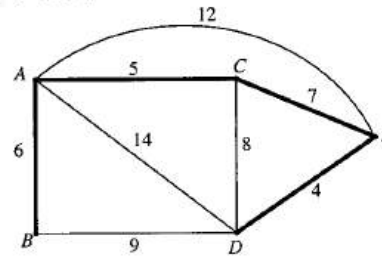


Answer B



Answer D

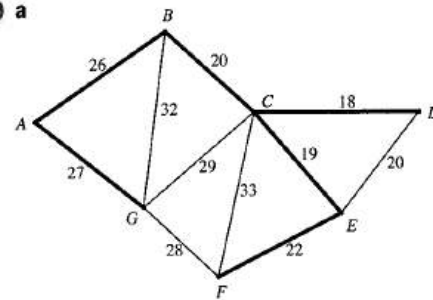
- 16 a and b



$6 + 5 + 7 + 4 = 22$

✓✓✓
 ✓
 (4 marks)

- 17 a



(1 mark)

b $27 + 26 + 20 + 18 + 19 + 22$
 $= 132$

The length of the minimum spanning tree is 132 km.

Total cost

$= \$26.50 \times 132 \times 1000$

$= \$3\,498\,000$

Profit

$= \$4\,100\,000 - \$3\,498\,000$

$= \$602\,000$

The company plans to make a profit of \$602 000.

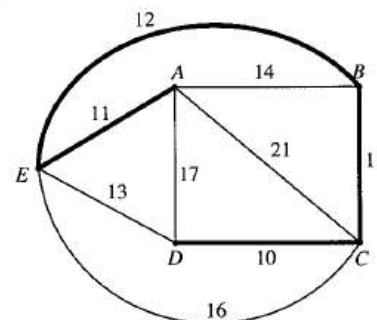
✓
 (2 marks)

18 $19 + 10 + 13 + 14 + 16 = 72$

The shortest distance is 72 km.

Answer C

- 19



$11 + 12 + 11 + 10 = 44$. The cost would be \$44 000.

✓✓✓
 ✓
 (4 marks)

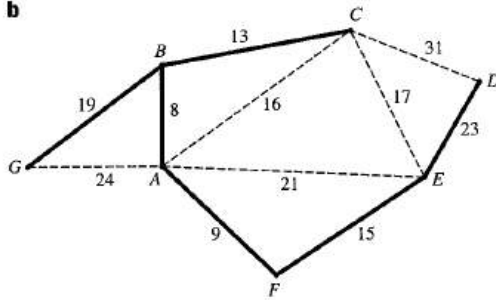
20 a $19 + 13 + 31 + 23 + 15 + 9 + 24 = 134$.

The distance is 134 m.

(1 mark)

Networking and Shortest Path

b



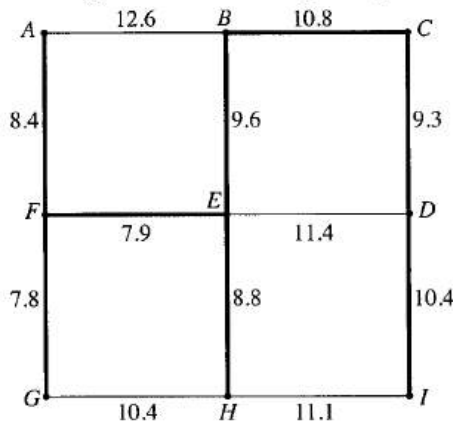
$19 + 8 + 13 + 9 + 15 + 23 = 87$ The distance is 87 m. (2 marks)

- 21 The path is $ACDEFG$.
As $17 + 16 + 17 + 19 + 14 = 83$, the minimum time is 83 minutes, or 1 h 23 min. (2 marks)

- 22 A spanning tree has one fewer edge than the number of vertices, which is $n - 1$.
Answer B

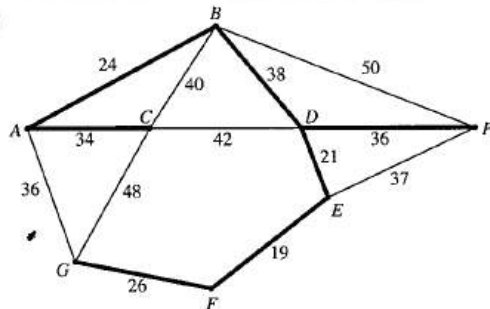
- 23 The 'shortest path' is $ADEF$.
As $16 + 7 + 9 = 32$, the cost of an adult ticket is \$32.
Total cost = $32 + 0.75 \times 32$
 $= 56$
The cost would be \$56. (2 marks)

- 24 $78 + 7.9 + 8.4 + 8.8 + 9.6 + 10.8 + 9.3 + 10.4 = 73.0$.
The length of the minimum spanning tree is 73 km.



Answer A

25



Minimum length
 $= 34 + 24 + 38 + 36 + 21 + 19 + 26$
 $= 198$
Cost = $\$18.40 \times 198$
 $= \$3643.20$ (3 marks)

- 26 Look for the vertices with odd degrees. He starts at C (degree 3) and finishes at I (degree 3).
As $9 \times 2 + 6 = 24$, there is a total of 24 km of tracks.
From I to C is 3 km.
Total distance = $24 + 3$
 $= 27$
Logan ran 27 km. (2 marks)